

Q:1 Image, imaging techniques, basic steps

1. Define Image

2. imaging techniques based on electromagnetic spectrum

3. basic steps of digital image processing

4. With block diagram, describe the approaches and components of image processing.

Define Image

An image is defined as a two-dimensional function, $F(x,y)$, where x and y are spatial coordinates, and the amplitude of F at any pair of coordinates (x,y) is called the **intensity** of that image at that point. When x,y and amplitude values of F are finite, we call it a **digital image**.

imaging techniques based on electromagnetic spectrum of light

and define each technique with its application areas

Imaging techniques are classified based on the **electromagnetic (EM) spectrum band** they use. Each band has a specific **wavelength, frequency, and energy**, which determines how it interacts with materials and what it is used for.

Gamma-Ray Imaging: Uses radioactive isotopes;
applications: PET image, bone scans, and reactor valve inspection.

X-ray Imaging: Radiates X-rays when electrons hit a nucleus.
applications : chest X-rays, angiograms, and CT scans

Ultraviolet (UV) Imaging: Uses UV radiation to excite high-energy electrons (fluorescence);
applications : used in microscopy for corn inspection.

Visible Imaging: Uses visible light (400–700 nm) to form images similar to human vision;
applications: photography, video recording, microscopy.

Infrared (IR) Imaging: Uses infrared radiation (>700 nm) to detect heat emitted by objects;
applications: thermal imaging, night vision, remote sensing

Microwave Imaging: Uses microwave radiation to penetrate objects
applications: Used in Radar; independent of weather conditions.

Radio Wave Imaging: Uses radio waves (very long wavelength, lowest energy) to capture signals from objects or distant sources;
applications: Used in MRI scans of the human body.

Low-Frequency Sound Wave Imaging: Uses low-frequency sound waves to create images of internal structures;
applications: underwater sonar, 3D map of ocean bed.

High-Frequency Sound Wave Imaging: Uses high-frequency sound waves for detailed imaging;
applications: medical ultrasonography, industrial non-destructive testing.

basic steps of digital image processing

Image Acquisition: This involves capturing the image using a sensor, such as a camera, scanner, or other device.

Image Enhancement: This step improves the visual quality of the image by adjusting brightness, contrast, and other properties.

Image Restoration: This stage aims to repair or recover degraded images, such as those affected by noise or blur.

Color Image Processing: This deals with manipulating color images, including color space conversions and color corrections.

Wavelets and Multiresolution Processing: This technique allows images to be represented at different resolution levels, enabling efficient analysis and processing.

Image Compression: This step reduces the storage space and transmission bandwidth of images while maintaining acceptable quality.

Morphological Processing: This involves manipulating image shapes and structures using set operations like erosion, dilation, opening, and closing.

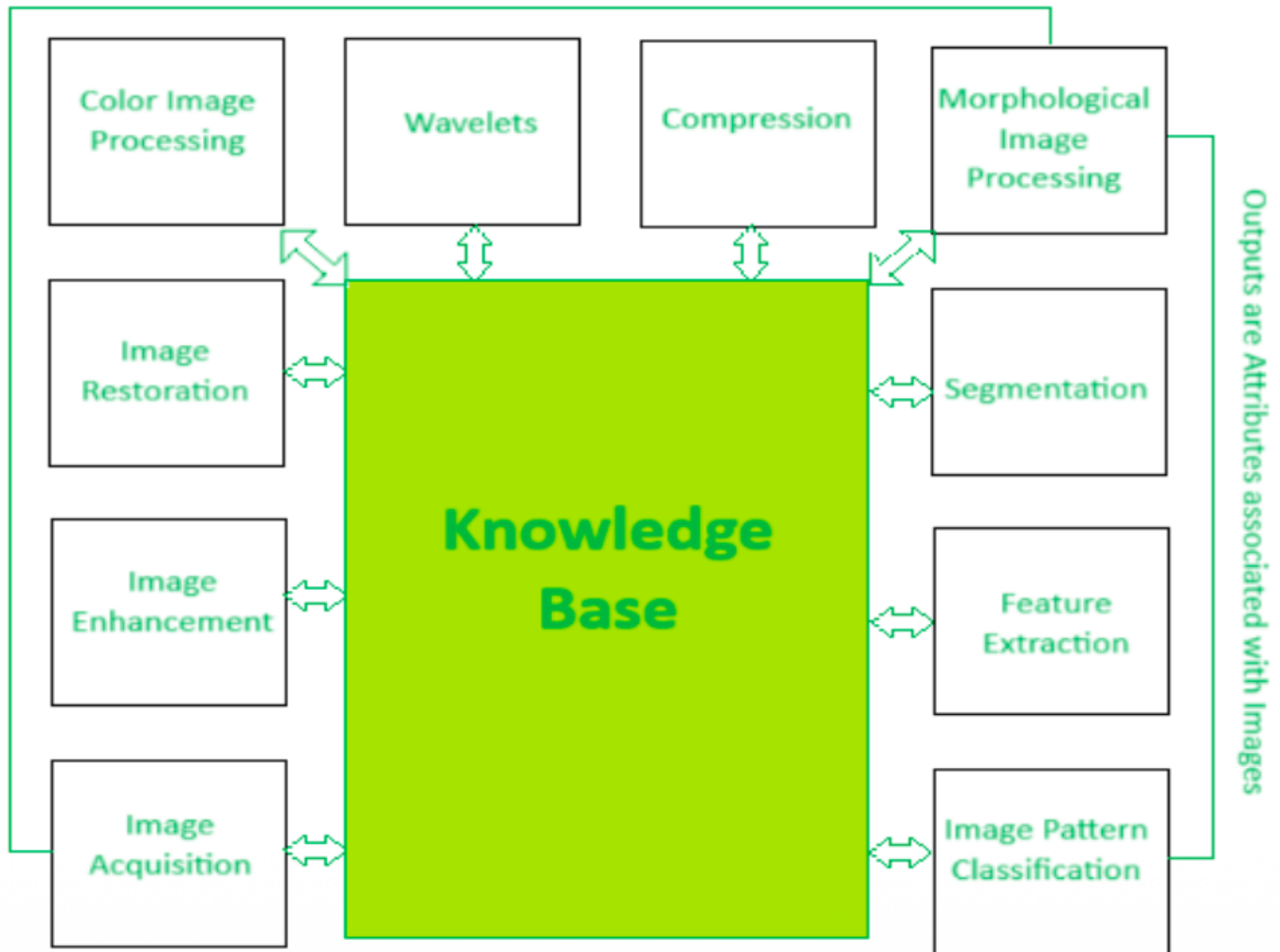
Image Segmentation: This stage separates an image into different regions based on intensity, color, or other characteristics.

Object Recognition: This involves identifying and classifying objects within an image based on their shape, texture, and other features.

Image Representation and Description: This stage focuses on transforming the image into a form suitable for computer processing, often involving boundary or region representation.

With block diagram, describe the approaches and components of image processing.

Outputs are Images





Components

